

# PROJECT SCOPING

**Project title:**

**Names of team members:**

Separated by semicolons

## 1. Project definition

Describe your project. What problem is your team addressing, why does it matter, and what solution, product, or deliverable do you plan to develop to address it?

Maximum 300 words

## 2. Project value proposition

What new knowledge, capability, improvement, or benefit will your project provide?

Maximum 100 words

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### 3. Deliverable type

Select all that apply.

- ☒ Physical prototype/product
- ☐ Research article
- ☐ Software application
- ☐ Computational model
- ☐ Data analysis or decision-support tool
- ☐ Design specification, CAD model, or design blueprint
- ☐ Educational resources or curriculum material
- ☐ Business plan/ proposal
- ☐ Community program/service
- ☐ Creative or media artifact
- ☐ Other (specify):

### 4. Stakeholder analysis

Identify the individuals, groups, or organizations that are affected by, influence, or have an interest in your project. Consider primary users, sponsors, customers, instructors, evaluators, community members, regulatory bodies, etc.

| Stakeholder | Needs/ Expectations | How will the project create value for this stakeholder? | Priority (High/Medium/Low) |
|-------------|---------------------|---------------------------------------------------------|----------------------------|
|             |                     |                                                         |                            |
|             |                     |                                                         |                            |
|             |                     |                                                         |                            |
|             |                     |                                                         |                            |
|             |                     |                                                         |                            |

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### 5. Technical constraints

List technical and engineering constraints associated with your project (performance requirements, design requirements, safety requirements, testing requirements, etc.).

- a)
- b)
- c)
- d)
- e)

### 6. External constraints

List external factors that constrain your project (e.g., government regulations, sponsor requirements, project deadlines, competition rules, etc.).

- a)
- b)
- c)
- d)
- e)

### 7. Internal constraints

List internal factors that constrain your project (e.g., budget, team expertise, available time, equipment availability, purchasing restrictions, workspace limitations, etc.).

- a)
- b)
- c)
- d)

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8. Trade-offs

What trade-offs is your team likely to encounter when making project decisions (e.g., cost vs. performance, speed vs. accuracy, durability vs. weight, etc.)?

a)

b)

c)

d)

9. Broader impacts

What broader impacts should be considered when developing this project? Consider social, environmental, economic, ethical, accessibility, health, safety, or sustainability impacts.

| Category | Impact |
|----------|--------|
|          |        |
|          |        |
|          |        |
|          |        |
|          |        |

10. Project risks

List potential risks that could negatively affect project success. Describe the potential impact of each risk and how your team plans to mitigate it.

| Risk | Potential impact | Mitigation strategy |
|------|------------------|---------------------|
|      |                  |                     |
|      |                  |                     |
|      |                  |                     |
|      |                  |                     |

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### 11. Success criteria

How will your team and stakeholders know the project is successful? Identify the criteria that will be used to evaluate project success. For each criterion, define what performance at the A, B, C, D, and F level would look like.

The grade levels represent your team's definition of project performance for each criterion and may be used during project evaluation.

Guidelines:

- i. Criteria should be SMART (Specific, Measurable, Achievable, Relevant, and Time-bound) and should reflect the project's objectives, stakeholder needs, and expected outcomes.
- ii. Criteria may relate to performance, functionality, safety, usability, sustainability, cost, documentation, stakeholder satisfaction, or other project objectives.
- iii. Define the performance levels (A, B, C, D, and F) for each criterion.

| Criterion | Grade |   |   |   |   |
|-----------|-------|---|---|---|---|
|           | A     | B | C | D | F |
|           |       |   |   |   |   |
|           |       |   |   |   |   |
|           |       |   |   |   |   |
|           |       |   |   |   |   |
|           |       |   |   |   |   |

## PROJECT SCOPING

### 12. Project timeline

Attach a project timeline in the form of a Gantt chart.

Guidelines:

- i. Include all major project tasks, milestones, and deliverables.
- ii. Show the planned start and completion dates (or weeks) for each task.
- iii. Ensure the timeline aligns with the project objectives, constraints, and success criteria.
- iv. Update the timeline throughout the project as needed